

14

C

A

$$v = \pm \omega \sqrt{x_0^2 - x^2}$$

$$\frac{30}{20} = \frac{\sqrt{x_0^2 - 20^2}}{\sqrt{x_0^2 - 30^2}}$$

$$x_0^2 = 1300$$

$$\frac{v}{30} = \frac{\sqrt{x_0^2 - 25^2}}{\sqrt{x_0^2 - 20^2}} = \frac{\sqrt{675}}{\sqrt{900}} = 0.866$$

$$v = 0.866 \times 30 = 26 \text{ mm s}^{-1}$$

15

A

x = A^2

A